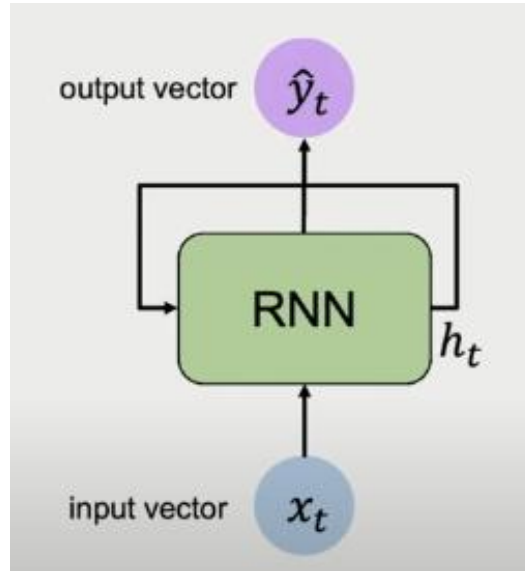


6CS012 – Artificial Intelligence and Machine Learning



Deep Sequence Models

Week 09 Lecture
Sunita Parajuli

Examples of sequence data

Speech recognition



“The quick brown fox jumped
over the lazy dog.”

Music generation

∅



Sentiment classification

“There is nothing to like
in this movie.”



DNA sequence analysis

AGCCCCTGTGAGGAACTAG



AGCCCCTGTGAGGAACTAG

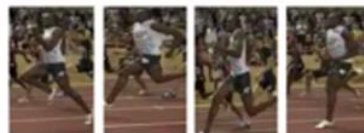
Machine translation

Voulez-vous chanter avec
moi?



Do you want to sing with
me?

Video activity recognition



Running

Name entity recognition

Yesterday, Harry Potter
met Hermione Granger.



Yesterday, **Harry Potter**
met **Hermione Granger**.

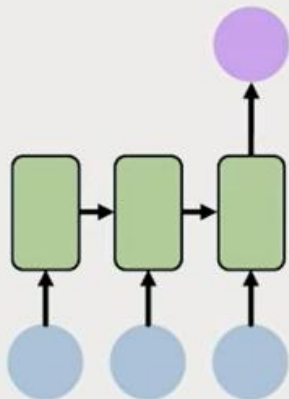
Sequence Modeling : The idea



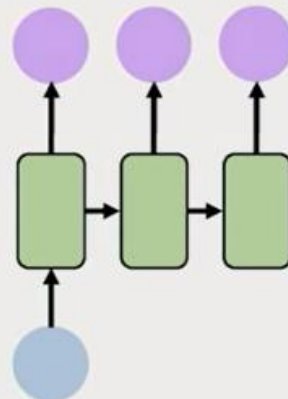
One to One
Binary Classification



“Will I pass this class?”
Student → Pass?



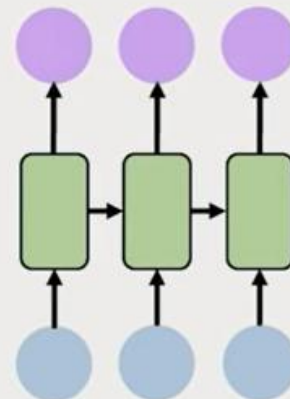
Many to One
Sentiment Classification



One to Many
Image Captioning



“A baseball player throws a ball.”

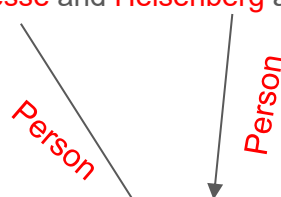


Many to Many
Machine Translation



An Example: Token classification

$x^{<1>}$ $x^{<2>}$ $x^{<3>}$ $x^{<4>}$ $x^{<5>}$ $x^{<6>}$
 $x = \text{Jesse and Heisenberg are good chef.}$



$y = 1\ 0\ 1\ 0\ 0\ 0$

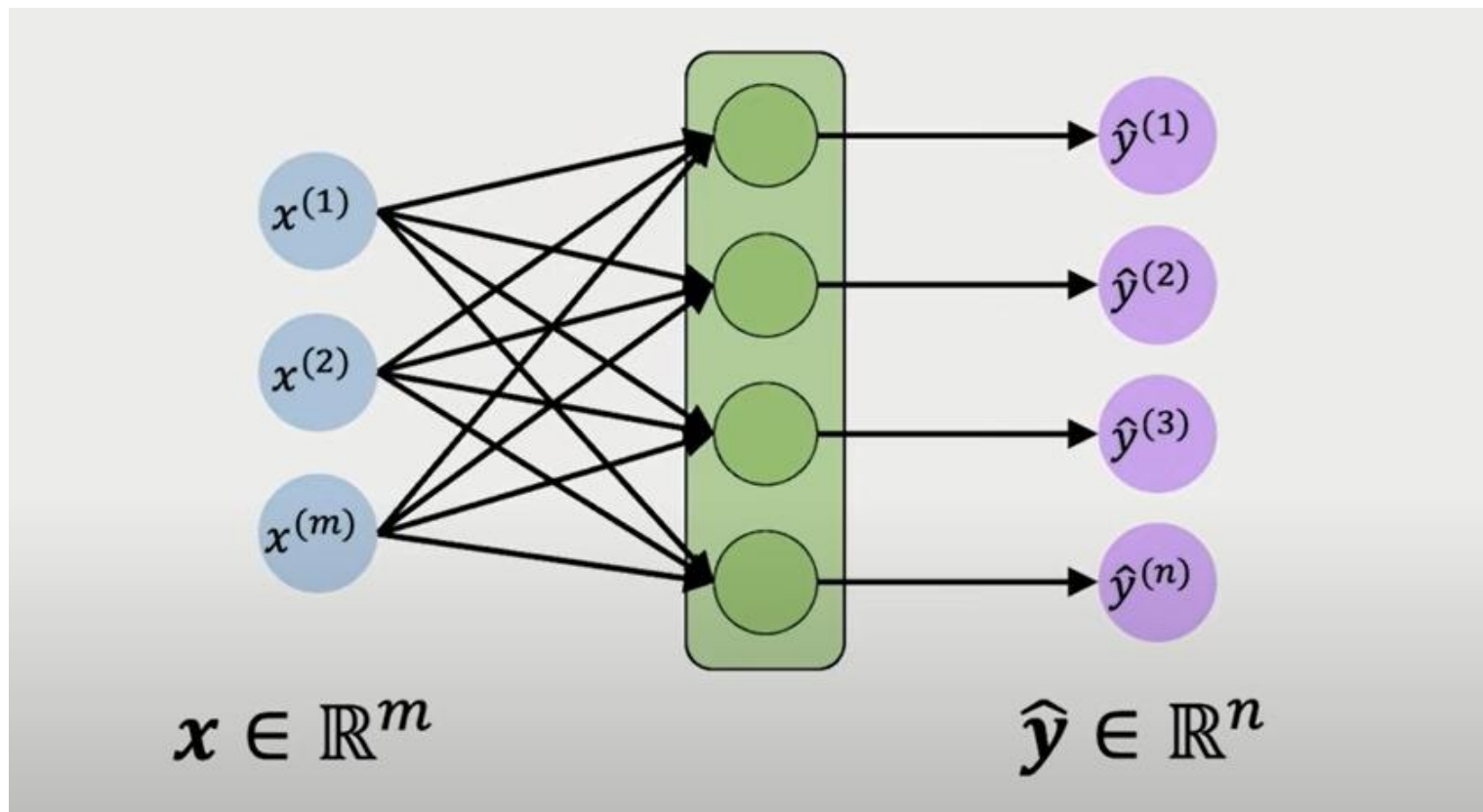
$y^{<1>} y^{<2>} y^{<3>} y^{<4>} y^{<5>} y^{<6>}$

English vocabulary

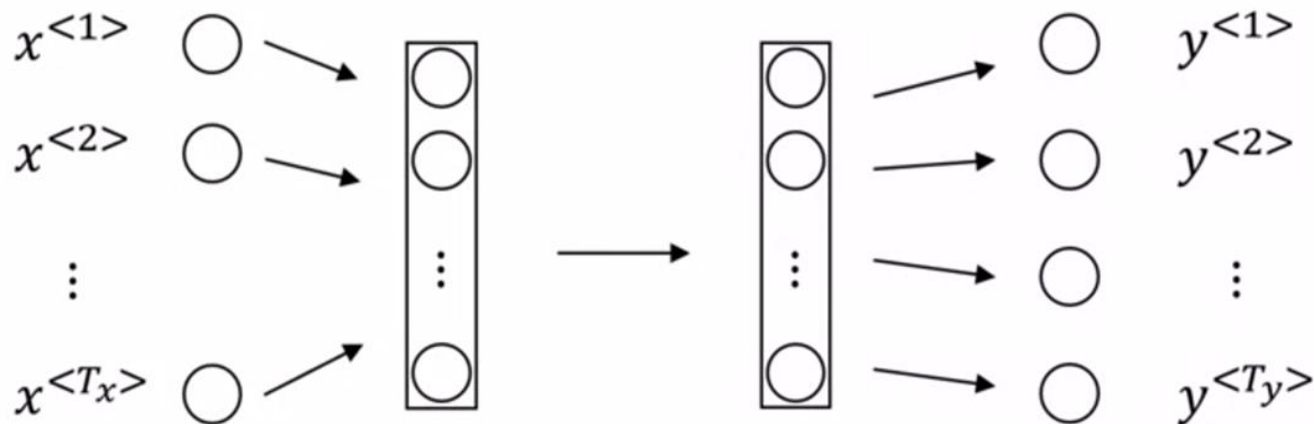
A	1
Aaron	2
...	...
...	...
Heisenberg	3456
...	...
Jesse	6754
...	...
Zulu	10000

	0		0
	0		...
$x^{<1>}$	...	$x^{<2>}$	1
=	1	=	...
	...		0
	0		0
	0		0

Vanilla Neural Network



Why not a standard network?



The food was
good, not bad at
all!

vs.

The food was bad,
not good at all!

Problems :

- Does not handle the variable length sequences
- Does not maintain information of order
- Does not share parameter across the sequence

Recurrent Neural Networks

Feed Forward Neural Network

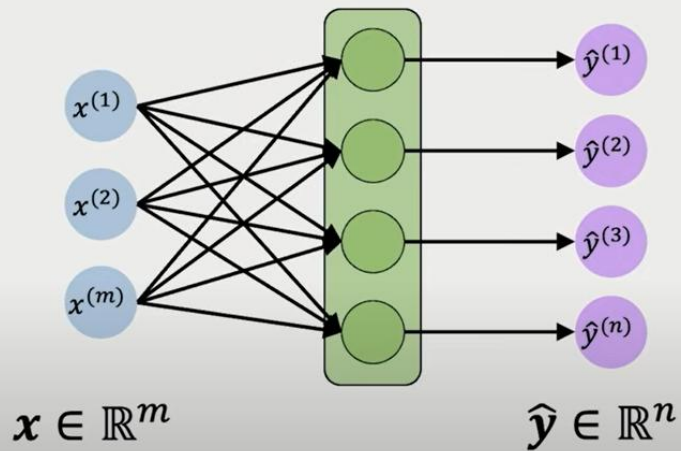


Image from MIT 6.S19 Deep Learning.

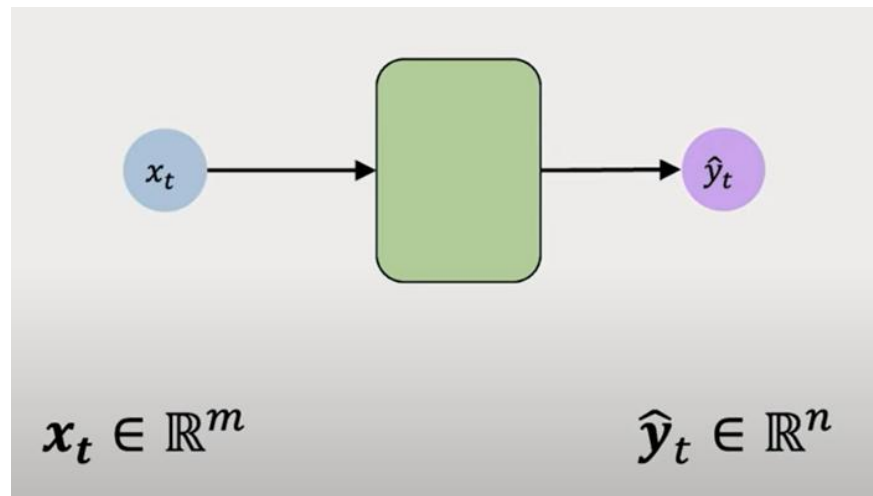
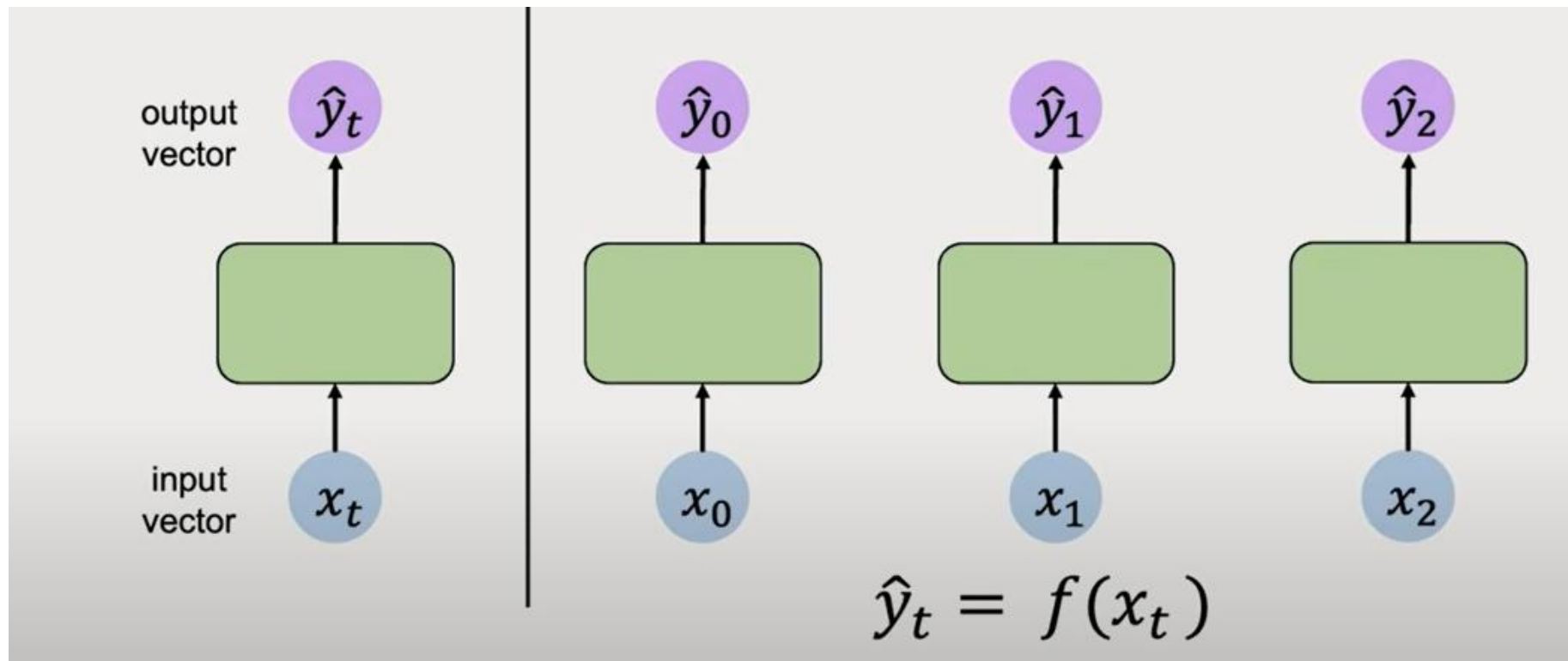
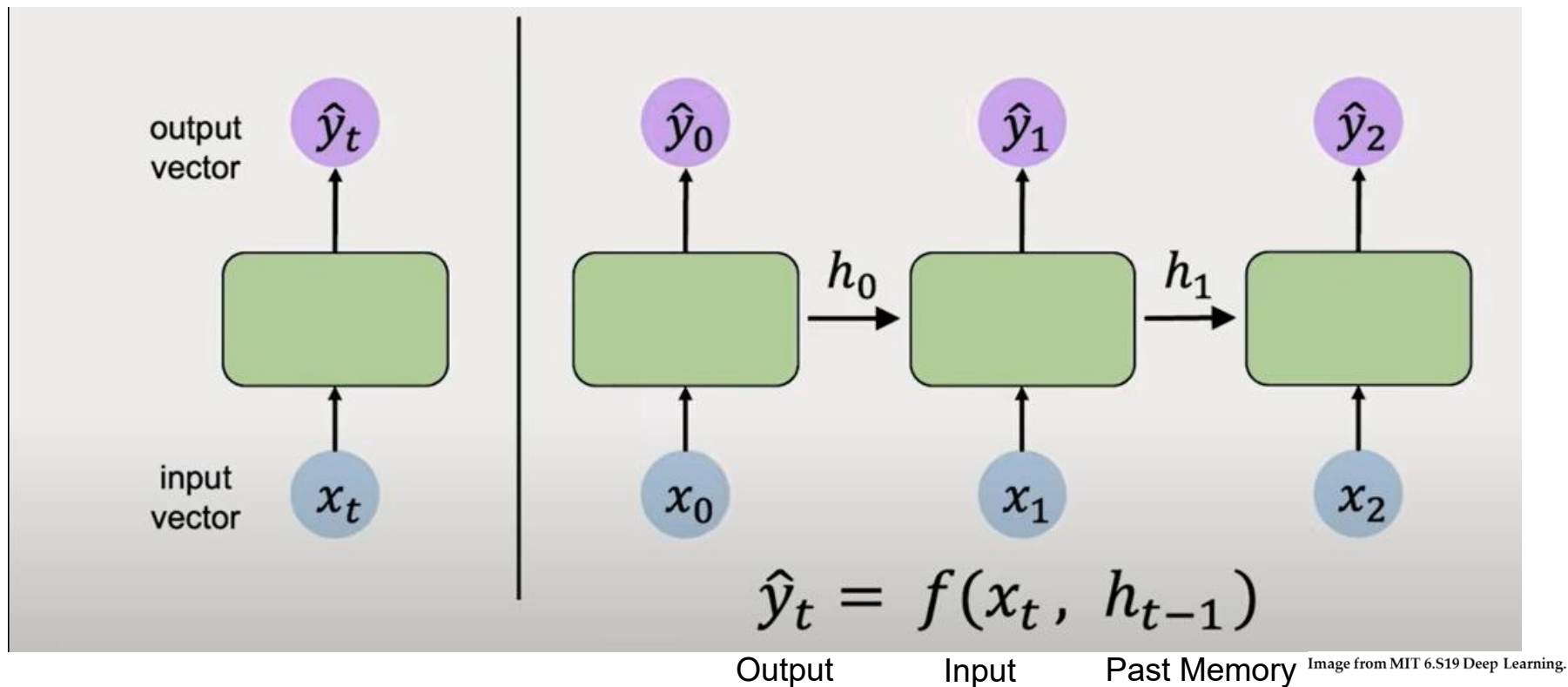


Image from MIT 6.S19 Deep Learning.

Handling Individual Time steps



Neuron with Recurrence



Recurrent Neural Networks (RNN)

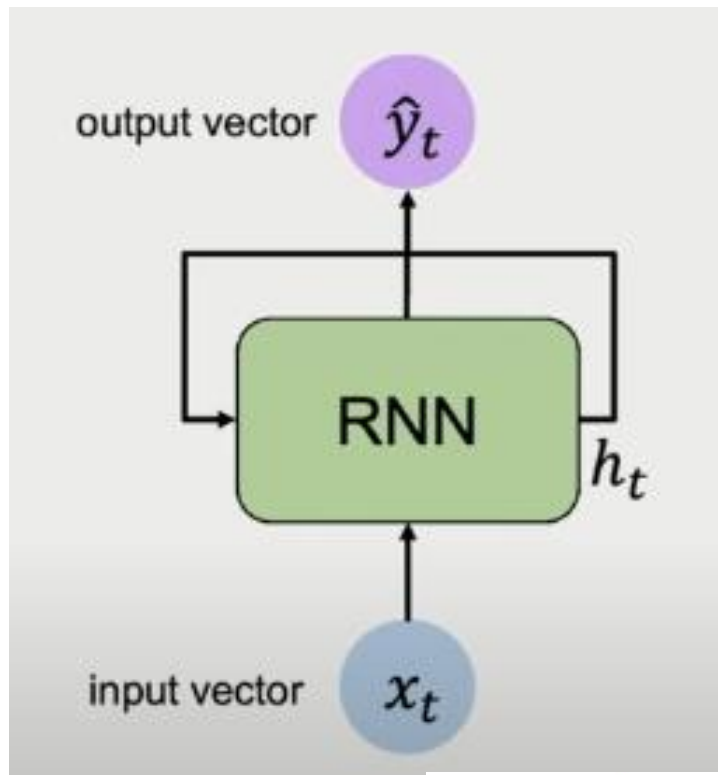


Image from MIT 6.S19 Deep Learning.

Apply a recurrence relation at every time step to process a sequence

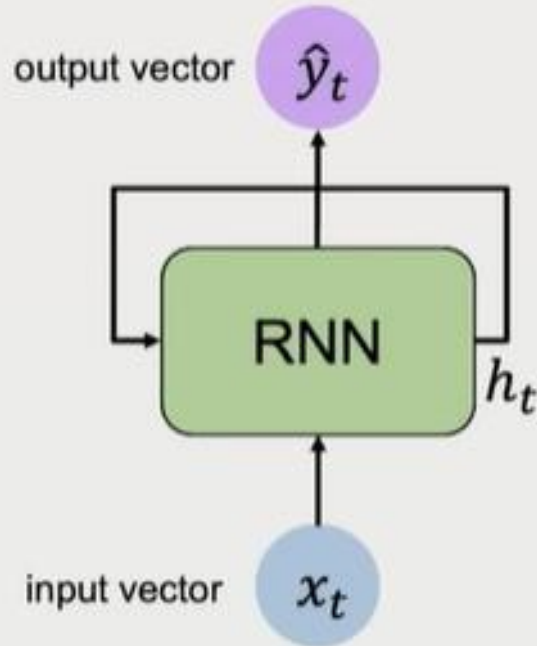
$$\boxed{h_t} = \boxed{f_W}(\boxed{x_t}, \boxed{h_{t-1}})$$

cell state function with weights W input old state

Note: the same function and set of parameters are used at every time step

RNNs have a state h_t that is updated at each time step as a sequence is processed

RNN state update and Output



Output Vector

$$\hat{y}_t = \mathbf{W}_{hy}^T h_t$$

Update Hidden State

$$h_t = \tanh(\mathbf{W}_{hh}^T h_{t-1} + \mathbf{W}_{xh}^T x_t)$$

Input Vector

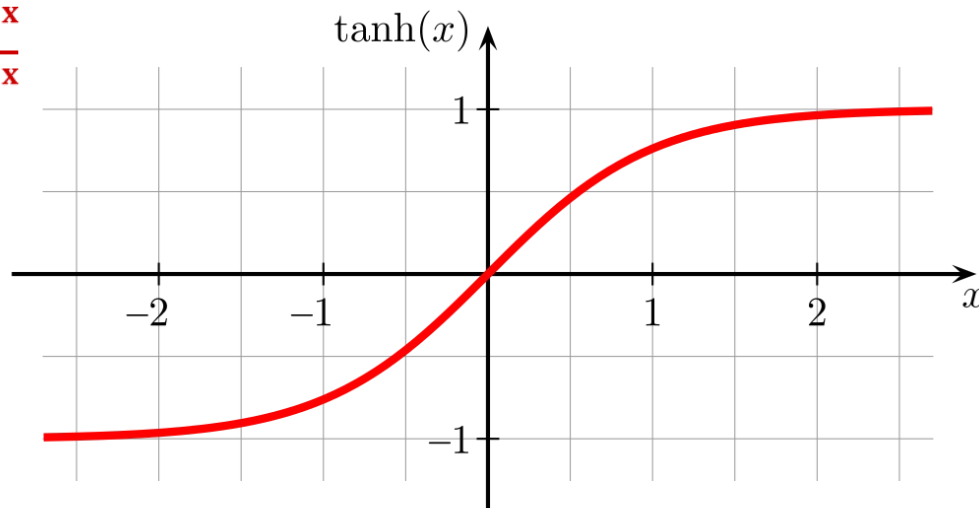
$$x_t$$

Tanh Activation Function

- Mathematical Representations:

- $\mathbf{f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}}$

- Output range: $(-1, 1)$



RNN Unrolled in time

The weight parameters are shared at every time step

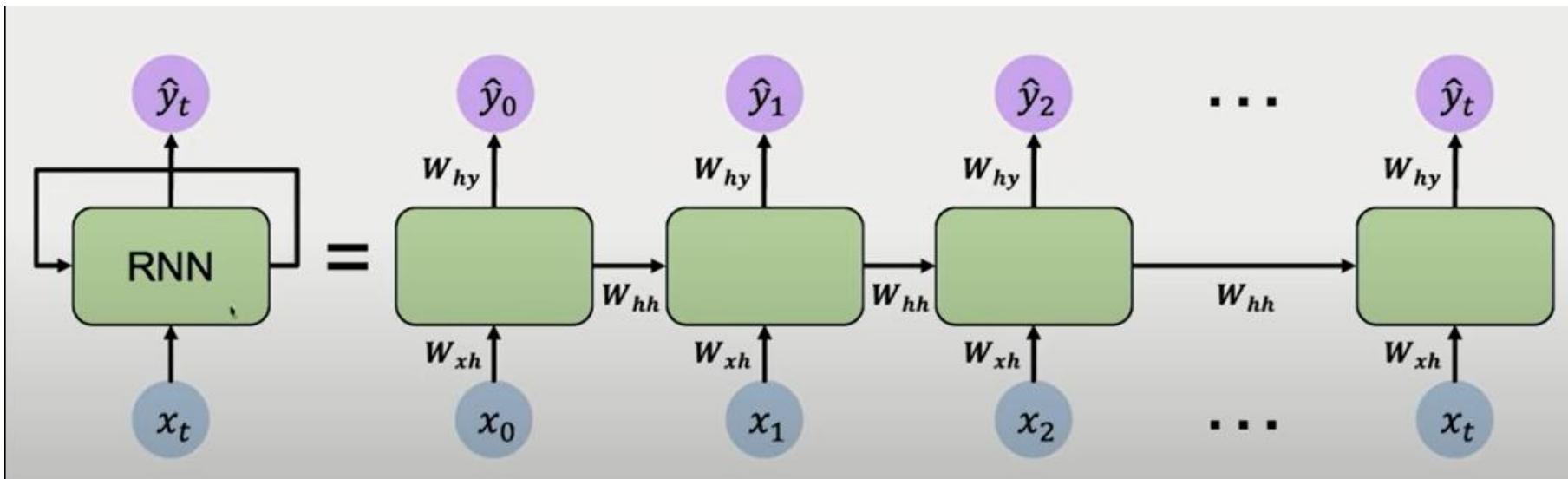


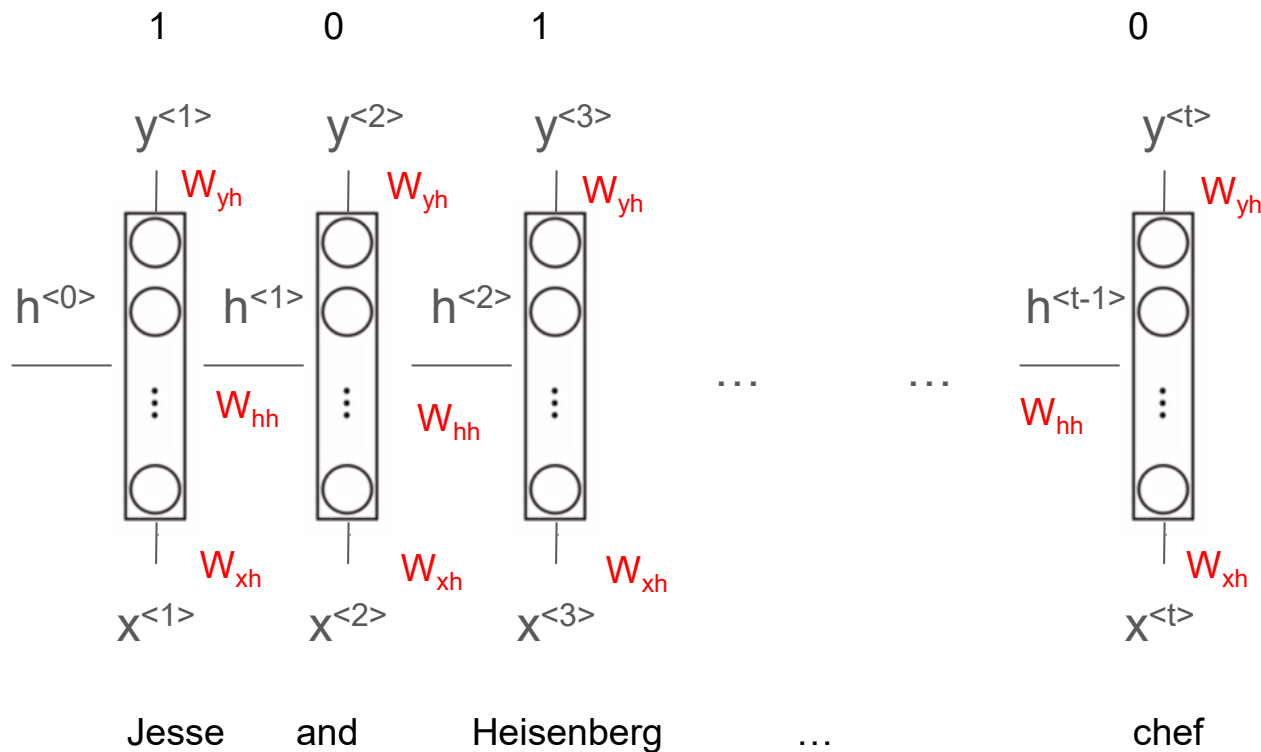
Image from MIT 6.S19 Deep Learning.

Frodo Baggins carried the One Ring from Rivendell to Mordor, aided by Samwise!

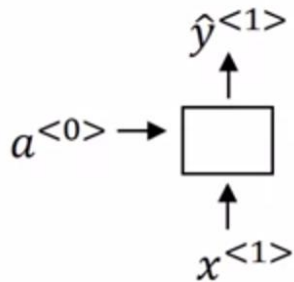
RNN model

$$h^{(t)} = \tanh(\overline{W}_{hh}h^{(t-1)} + \overline{W}_{xh}x^{(t)} + b_h)$$

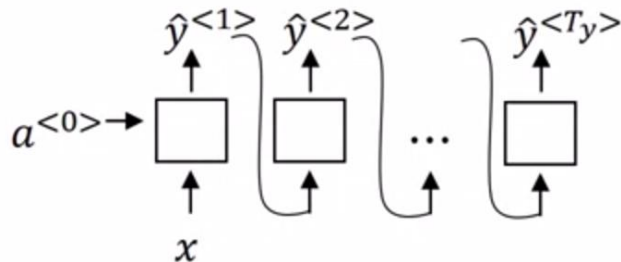
$$y^{(t)} = g(\overline{W}_{yh}h^{(t)} + b_y)$$



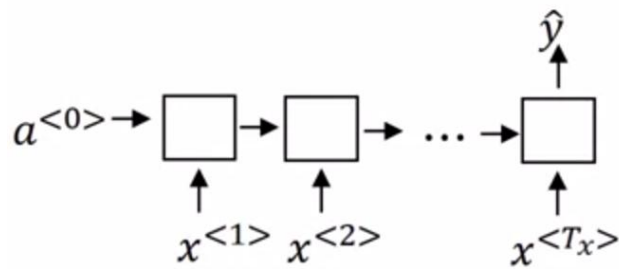
Types of RNN



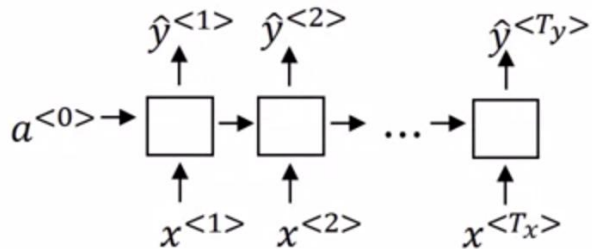
One to one



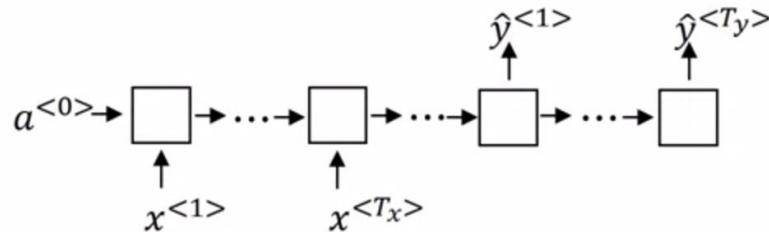
One to many



Many to one



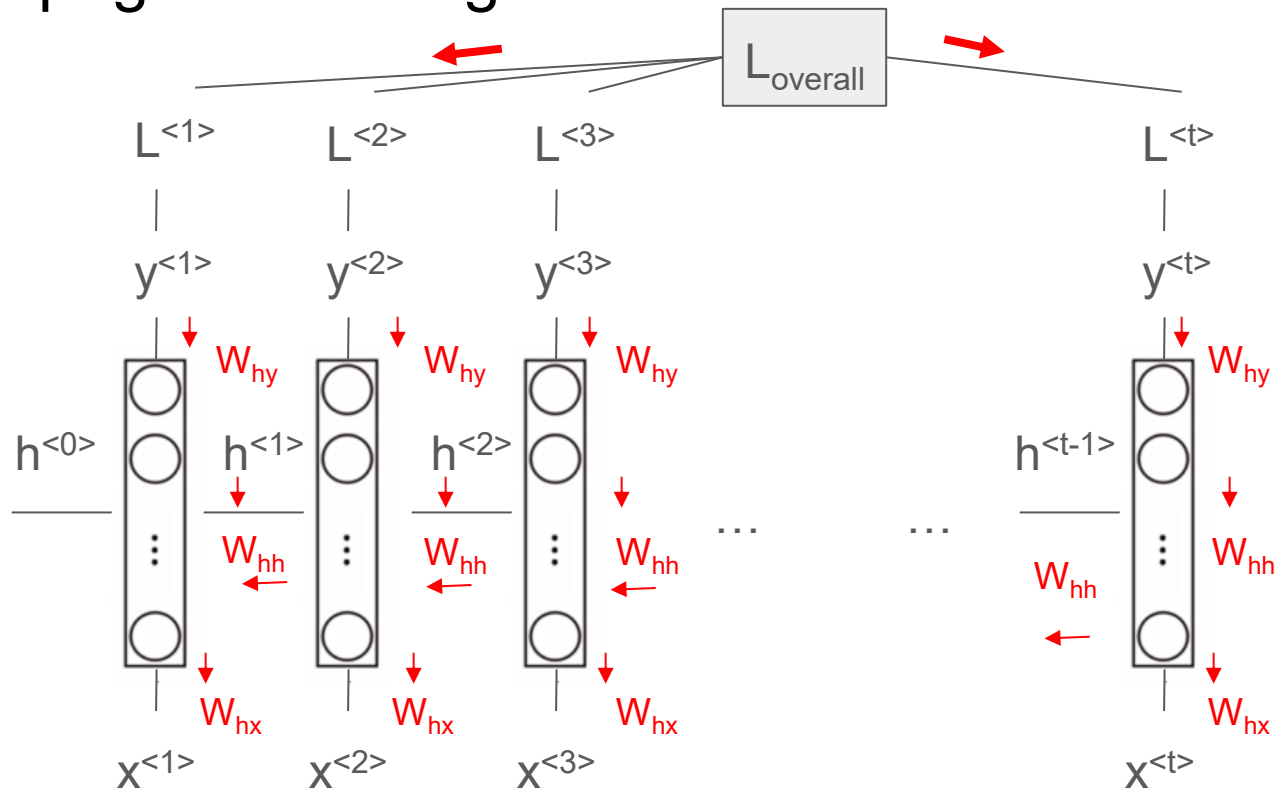
Many to many



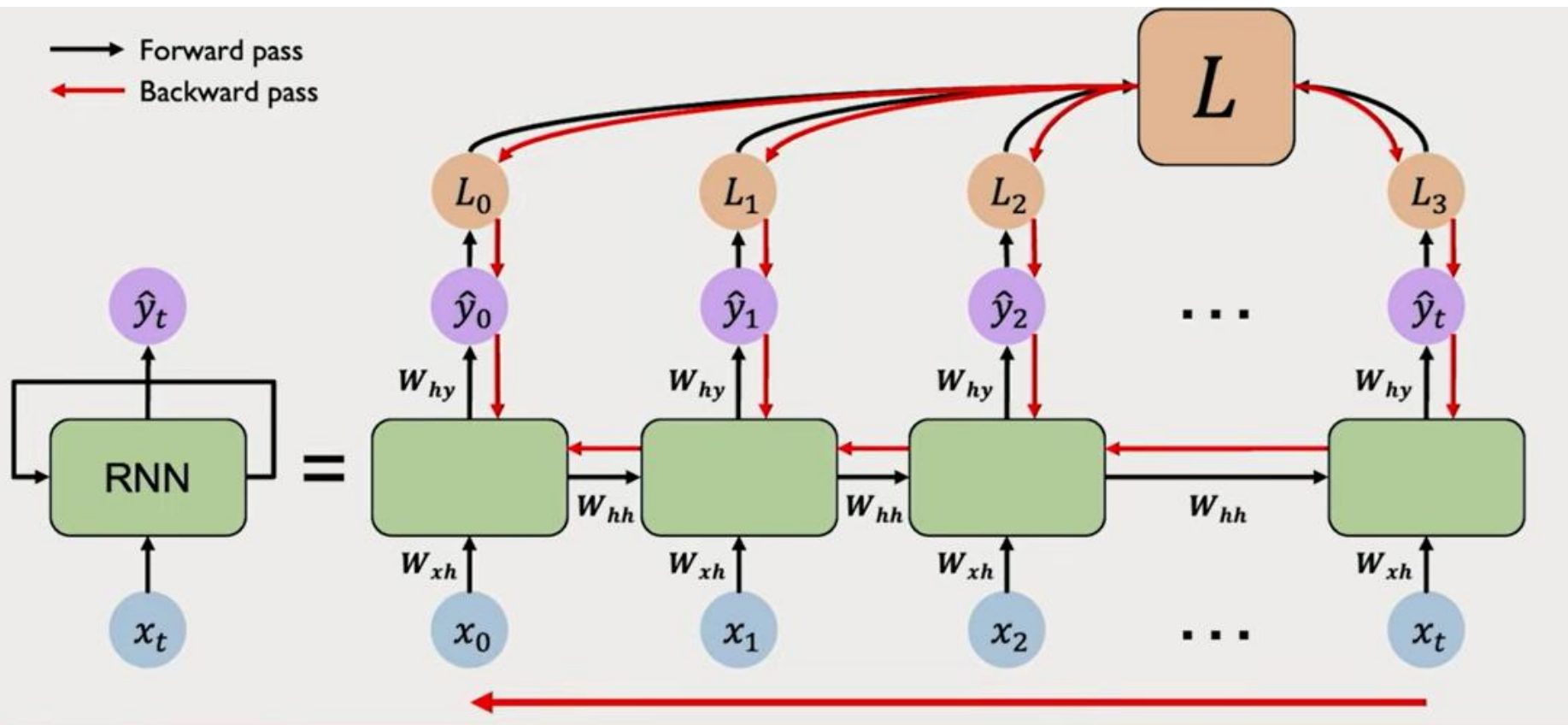
Many to many

Training RNNs

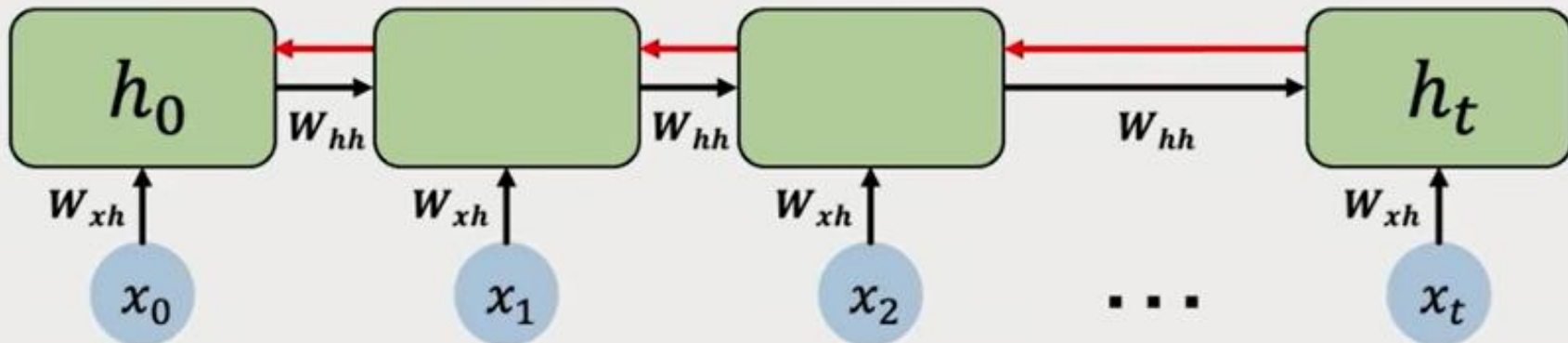
Backpropagation through time



Backpropagation Through Time (Training RNNs)



RNN Gradient Flow and Vanishing Gradient Problem



Computing the gradient with respect to h_0 involves many factors of w_{hh} in addition to repeated gradient computation.

If many values are greater than 1 → **Exploding gradients:**

- *Gradient clipping* to scale down large gradients
- Weight Initialization

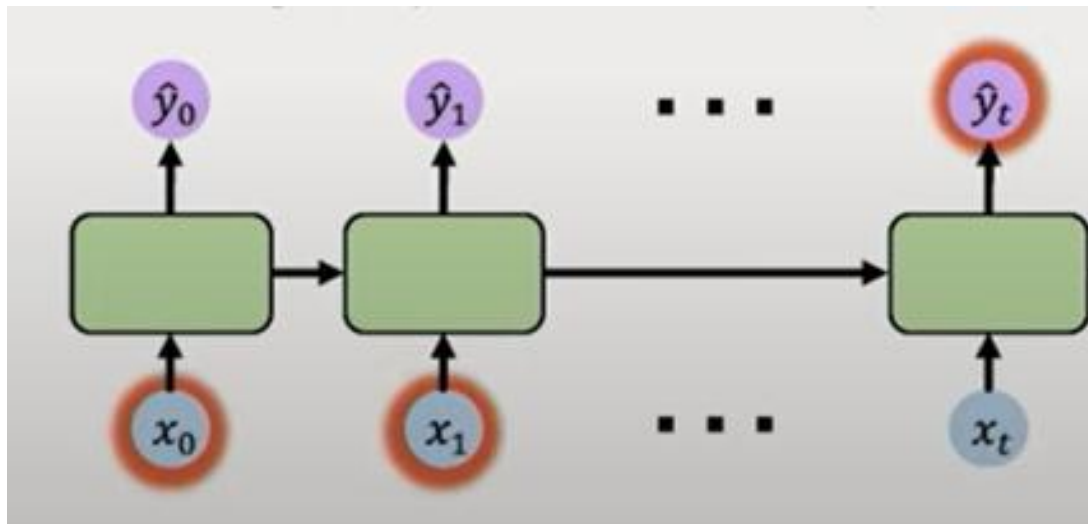
If many values are less than 1 → **Vanishing gradients**

- Activation function
- Weight initialization
- Network architecture

Long Term Dependency Problems

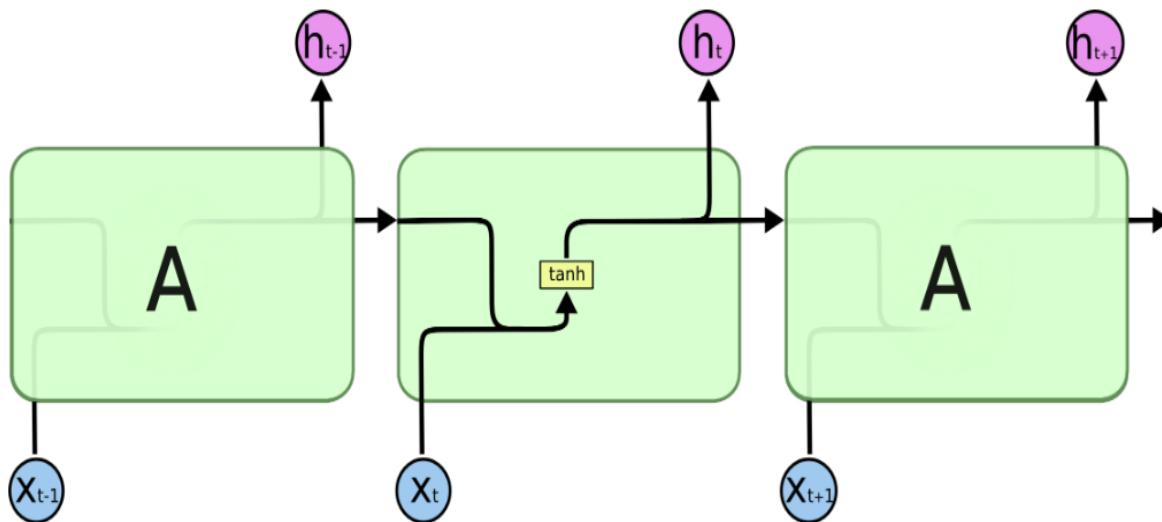
Consider the example below !

I watched a movie called Interstellar yesterday. It had amazing visuals and explored the concept of time dilation in space. I loved the movie, maybe I should explore more into _____ genre.



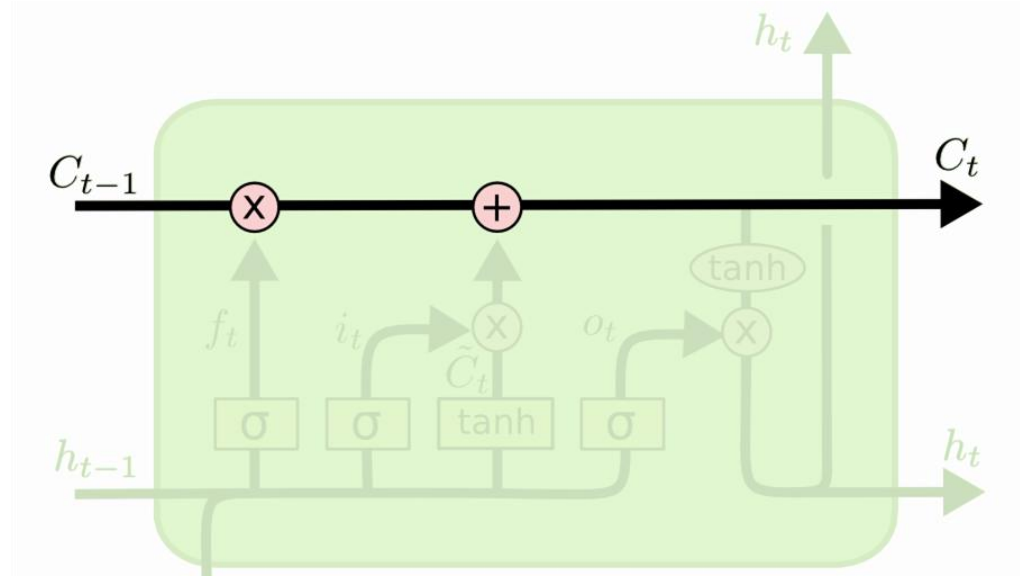
Long Short Term Memory (LSTM)

RNN Models (simplified architecture)



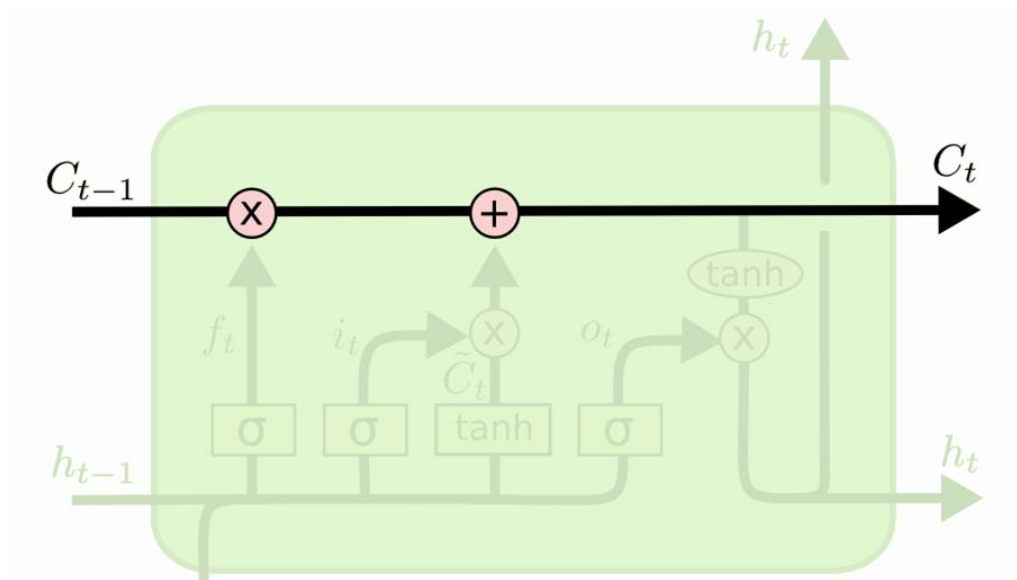
The repeating module in a standard RNN contains a single layer.

Introducing Long term Memory



I watched a movie called *Interstellar* yesterday. It had *amazing visuals* and explored the concept of *time dilation in space*. I loved the movie , maybe I should explore more into _____ genre."

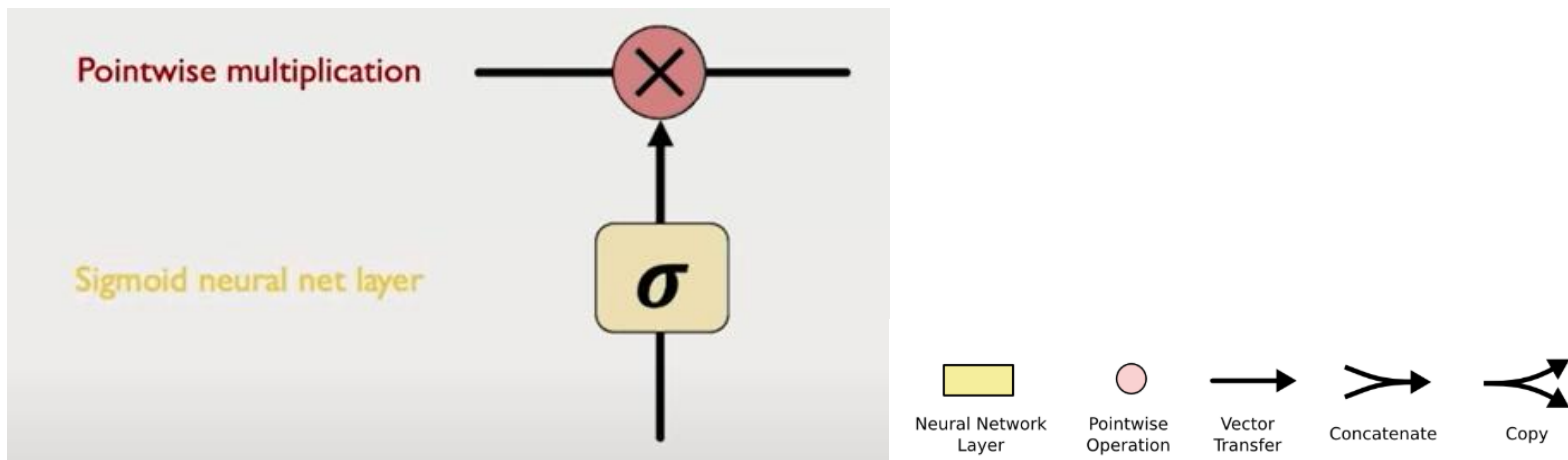
Introducing Long term Memory



I watched a movie called *Interstellar* yesterday. It had *amazing visuals* and explored the concept of *time dilation in space*. I loved the movie , maybe I should explore more into _____ genre. But my sister *hated* it . she *loves simple* and *heart-warming* moves about *love*. She is more into _____ genre.

LSTM Idea : Gating Mechanism!

Use gates to selectively add or remove information within each recurrent unit

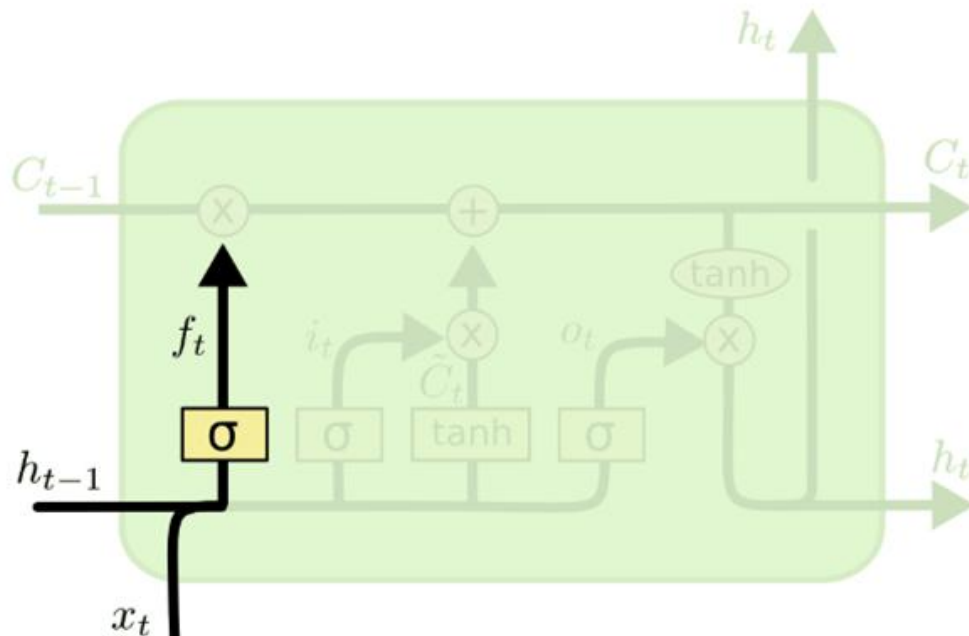


A **sigmoid activation layer**: outputs values between 0 and 1 describes how much of each component should be let through.

- A value of 0 means let nothing through
- A value of 1 means let everything through

Forget Gate

- Decides what information to discard from the previous cell state $\{C_{t-1}\}$
- **Input:**
 - Previous hidden state h_{t-1} and current input x_t .
- **Operation:**
$$\mathbf{f}_t = \sigma(\mathbf{W}_f \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f)$$
- The output $\mathbf{f}_t$ (values between 0 and 1) determines how much cell **state** C_{t-1} should be retained at current cell.



I watched a movie called *Interstellar* yesterday. It had *amazing visuals* and explored the concept of *time dilation in space*. I loved the movie , maybe I should explore more into _____ genre."

Input Gate

- **Purpose:**

- Decides what new information to store in the cell state.

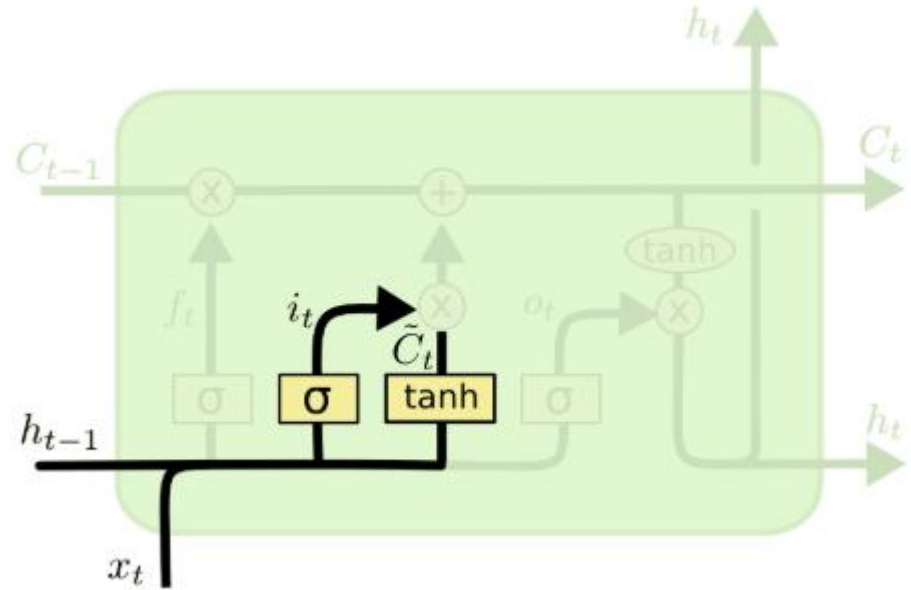
- **Operation:**

- Operation – 1 – Input gate layer (sigmoid):

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i)$$

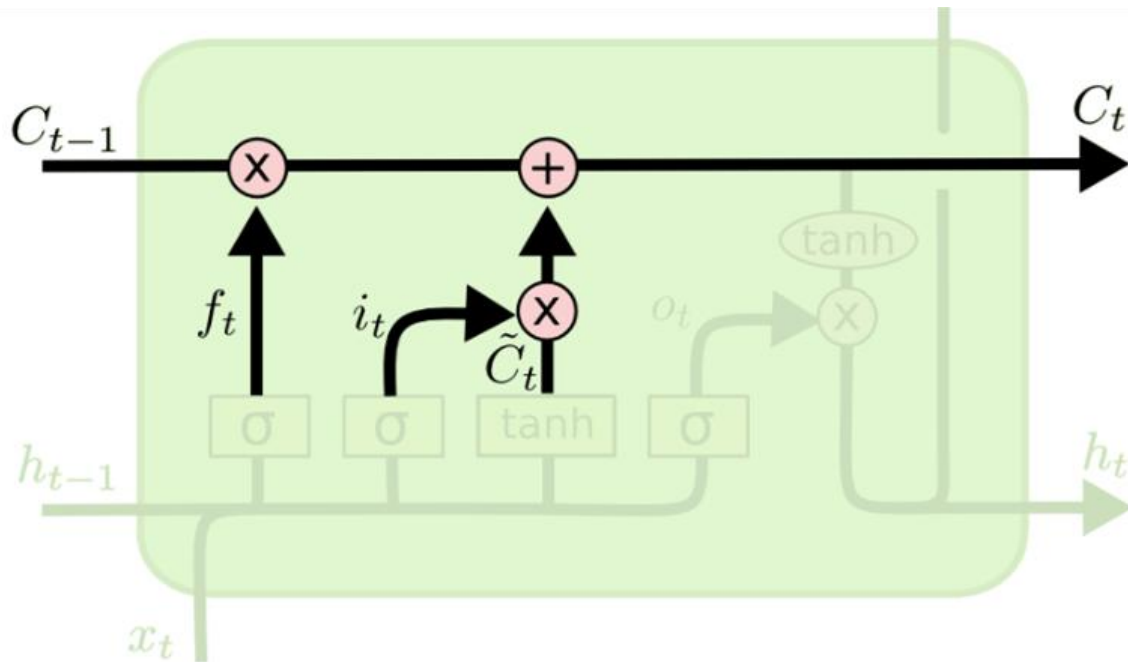
- Operation – 2 – Candidate values:

- $\tilde{\mathbf{C}}_t = \tanh(\mathbf{W}_c \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c)$



I watched a movie called *Interstellar* yesterday. It had *amazing visuals* and explored the concept of *time dilation in space*. I loved the movie , maybe I should explore more into _____ genre."

Remember: Gated Operations in forget and input gate!



Output Gate

Purpose:

- Determines the next hidden state h_t , which is also the output of the LSTM cell.

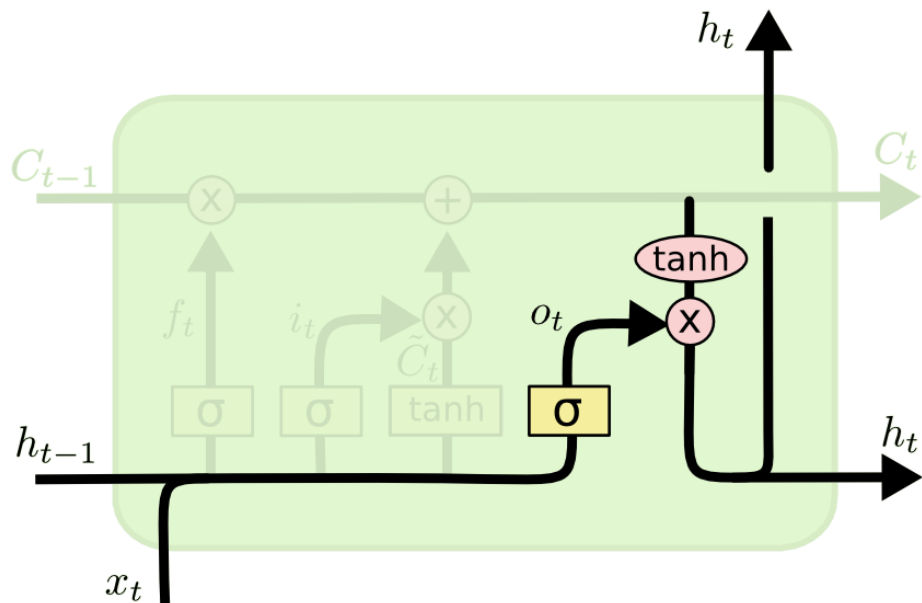
Operations:

- Operation – 1 – sigmoid layer:
 - Decides which parts of the cell state will influence the output.

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o)$$

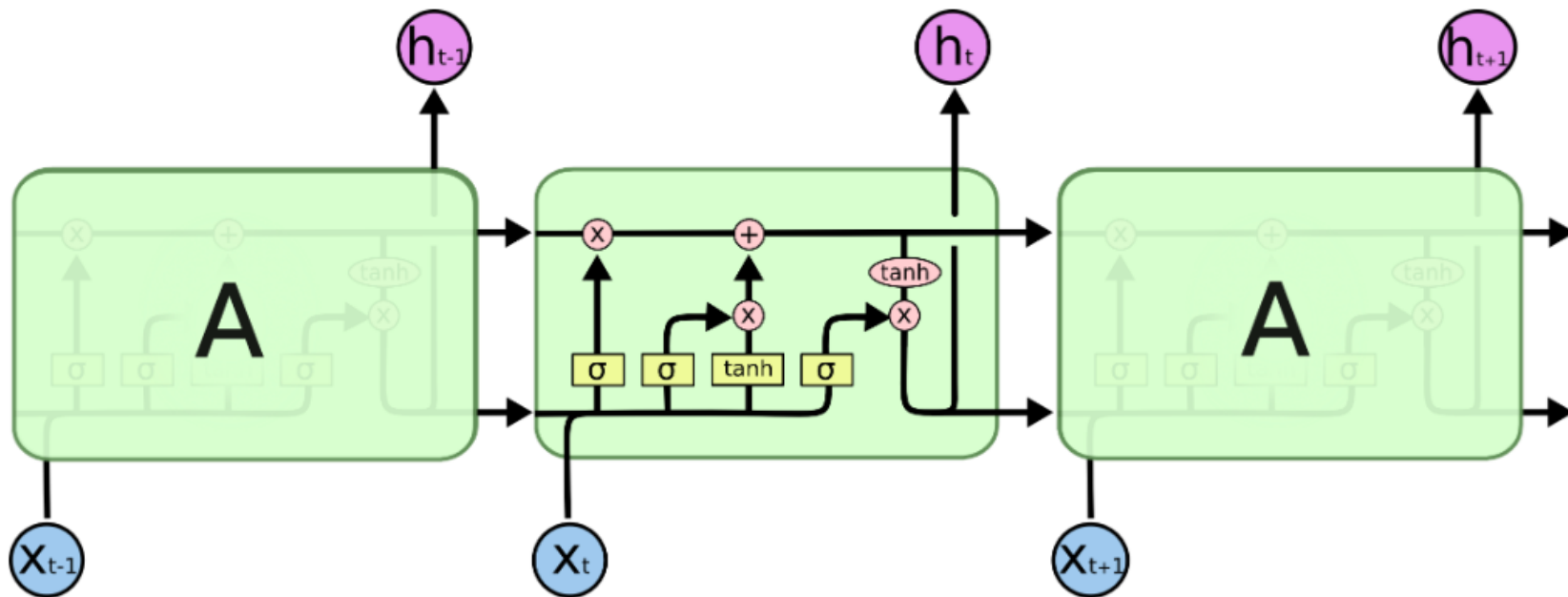
- Operation – 2 – Hidden state calculation:
 - Pass the updated cell state through tanh and multiply element wise **with** $\mathbf{o}_t$ to get the final **hidden state h_t** :

$$\mathbf{h}_t = \mathbf{o}_t * \tanh(\mathbf{C}_t)$$



I watched a movie called *Interstellar* yesterday. It had *amazing visuals* and explored the concept of *time dilation in space*. I loved the movie , maybe I should explore more into _____ genre."

LSTM Architecture (Summary)



The repeating module in an LSTM contains four interacting layers.

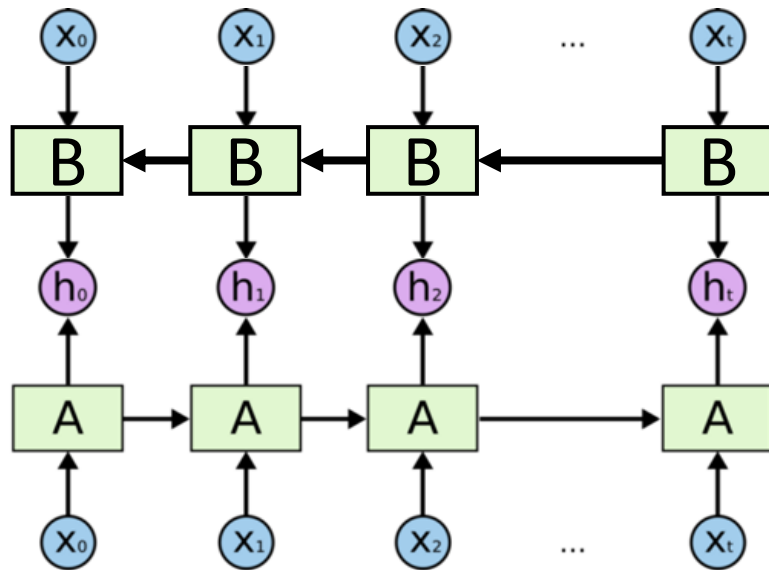
I watched a movie called *Interstellar* yesterday. It had *amazing visuals* and explored the concept of *time dilation in space*. I loved the movie , maybe I should explore more into _____ genre."

Bidirectional RNN

Consider the following Sentence:

I hate **Apple**. Their products are overpriced and overhyped.

What is Apple here?



Captures forward and backward information from the input sequence!!

Sequence to Sequence Models

Seq2Seq Models

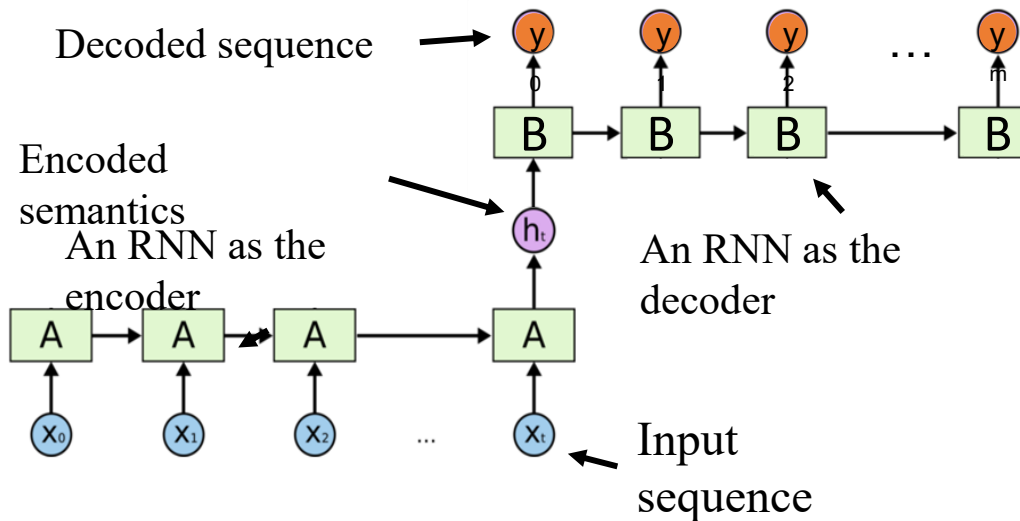
Encoder and one Decoder network

Encoder RNN: extract and compress the semantics from the input sequence

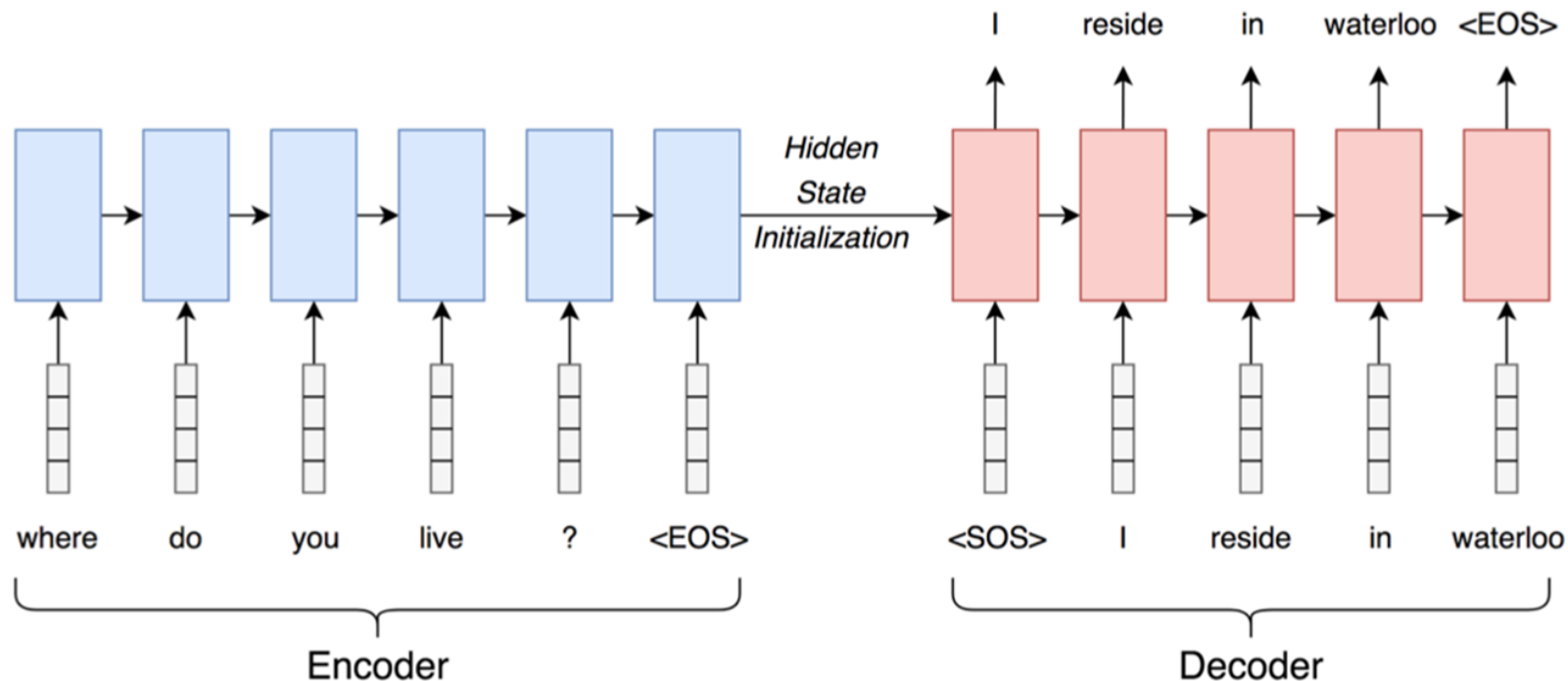
$h_o(src)$ and $C_0(src)$.

Decoder RNN: generate a sequence based on the input semantics

$h_o = LSTM -$
 $GRU_{\theta}(h_{o-1}, y_{o-1})$



Seq2Seq Models Application (Machine Translation)



References

- <https://colah.github.io/posts/2015-08-Understanding-LSTMs/>
- <https://introtodeeplearning.com/>

Thank You!